

UDT

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Brute idea for coherent large beliefs

Suppose we have "good guesses" $\mathbb{P}(\varphi)$ for all $|\varphi| \leq n$.
(Ignore for now that this is still an infinite set if there are infinite atoms.)
(Think about doing Gödel numbering instead.)

How can we assign a good guess to $|\psi| > n$?

Well, we can express ψ as the composition of (a small number of) formulas $\leq n$:

$$\begin{aligned}\psi &= \varphi(\varphi_1, \dots, \varphi_m) \\ \varphi_i &= \varphi^i(\varphi_1^i, \dots, \varphi_k^i) \\ &\quad \text{etc.}\end{aligned}$$

A static program can find this expression automatically.

If the sentence is so big that this takes too long, then allegedly we don't even need to assign a probability to ψ ?

If we only know the probabilities of sentences, we can't do much. We can do something like $\mathbb{P}(\psi) = \mathbb{P}(\forall \bar{x} \varphi)$ (since maybe the probability of $\forall \bar{x} \varphi$ is a good first guess at the proportion of valuations that make φ true).

But if we were somehow tracking not only probabilities of sentences, but of formulas under different probabilistic valuations, then we could do more. That is, suppose for every $|\varphi| \leq n$ we have a good guess for the function $f_\varphi : \mathbb{Q}^{|\bar{x}|} \rightarrow \mathbb{Q} \cap [0, 1]$ defined $f_\varphi(x_1, \dots, x_m) := \mathbb{P}(\varphi(\varphi_1, \dots, \varphi_m) \mid \mathbb{P}(\varphi_i) = x_i)$. (This is starting to look like just using many-valued logics.)

Then we could iteratively compute $\mathbb{P}(\psi)$.

This is a simplified (less computationally expensive, and losing more information I think? Now I'm not sure) version of doing $\mathbb{P}(\psi) = \sum_{s \in \{0,1\}^n} \mathbb{P}(\varphi(s_1, \dots, s_n)) \prod_{1 \leq i \leq n} \mathbb{P}(\varphi_i = s_i)$.

I'm not sure how much this differs from the above, since we still ignore correlations between the φ_i . Maybe due to some consistency constraints (which are approximately satisfied for small "good guesses") the two calculations should approximately coincide. I feel like this should be the case, I'll think about that.

Problems:

- Why should this coincide with "actually useful guesses" or "guesses actually having good consequences on average"?
- How do we get good guesses of f_φ ? The only thing we ever observe are binary statements about sentences being true or false. If we go fine-grained enough (latter version) and know the values of all atoms, we can just compute. If we lack the value of some atom, which probability do we assign to it?
- If atoms were just "propositional variables", maybe 0.5 is the sane initial guess (Bayesian updated on known sentences involving it). But we're interested in prime formulas as atoms. Is this, assigning good first guesses at quantified formulas, the hard core of the problem? Yes, because we face infinities? (Paul is thinking about this.)
- Probably this will present problems similar to ARC's cumulant propagation because of missing higher-order correlations (that could in principle be arbitrarily bad)?
- Concerns about computation and memory, but maybe we could use dynamic programming and obtain guesses of f_φ for big φ by polynomial extrapolation? And again, why should this correspond to the notion we're actually searching for (except when we have some guarantees on the correctness of the fit due to small-degree polynomials)?